INNER JOIN

Employee			
Eid	Name	Salary	**Dept_No**
101	Sanika	23000	1
102	Pranoti	34000	2
103	Lokesh	45000	NULL
104	Sanket	12000	2
105	Ashwini	23000	1
106	Seema	30000	NULL

Dept		
Dept_No	D_Name	Location
1	IT	Pune
2	Admin	Mumbai
3	HR	Pune
4	SCM	Mumbai

INNER JOIN: Fetch only matching records from both tables

After inner join:

```
SELECT * FROM employee INNER JOIN dept
ON employee.Dept_No = dept.Dept_No;
```

Eid	E_Name	Salary	Dept_No	Dept_No	D_Name	Location
101	Sanika	23000	1	1	IT	Pune
102	Pranoti	34000	2	2	Admin	Mumbai
104	Sanket	12000	2	2	Admin	Mumbai
105	Ashwini	23000	1	1	IT	Pune

LEFT OUTER JOIN

Employee			
Eid	Name	Salary	**Dept_No**
101	Sanika	23000	1
102	Pranoti	34000	2
103	Lokesh	45000	NULL
104	Sanket	12000	2
105	Ashwini	23000	1
106	Seema	30000	NULL

Dept		
Dept_No	D_Name	Location
1	IT	Pune
2	Admin	Mumbai
3	HR	Pune
4	SCM	Mumbai

LEFT OUTER JOIN: Matching records from both tables**,** non-matching records from left table with null values for right table

After left outer join:

```
SELECT * FROM employee LEFT JOIN dept
ON employee.Dept_No = dept.Dept_No;
```

Eid	E_Name	Salary	Dept_no	Dept_no	D_Name	Location
101	Sanika	23000	1	1	IT	Pune
102	Pranoti	34000	2	2	Admin	Mumbai
104	Sanket	12000	2	2	Admin	Mumbai
105	Ashwini	23000	1	1	IT	Pune
103	Lokesh	45000	NULL	NULL	NULL	NULL
106	Seema	30000	NULL	NULL	NULL	NULL

RIGHT OUTER JOIN

Employee			
Eid	Name	Salary	**Dept_No**
101	Sanika	23000	1
102	Pranoti	34000	2
103	Lokesh	45000	NULL
104	Sanket	12000	2
105	Ashwini	23000	1
106	Seema	30000	NULL

Dept		
Dept_No	D_Name	Location
1	IT	Pune
2	Admin	Mumbai
3	HR	Pune
4	SCM	Mumbai

RIGHT OUTER JOIN: Matching records from both tables**,** non matching records from right table with null values for left table

After right outer join:

```
SELECT * FROM employee RIGHT JOIN dept
ON employee.Dept_No = dept.Dept_No;
```

Eid	E_Name	Salary	Dept_No	Dept_No	Name	Location
101	Sanika	23000	1	1	IT	Pune
102	Pranoti	34000	2	2	Admin	Mumbai
104	Sanket	12000	2	2	Admin	Mumbai
105	Ashwini	23000	1	1	IT	Pune
NULL	NULL	NULL	NULL	3	HR	Pune
NULL	NULL	NULL	NULL	4	SCM	Mumbai

FULL OUTER JOIN

Employee			
Eid	Name	Salary	**Dept_No**
101	Sanika	23000	1
102	Pranoti	34000	2
103	Lokesh	45000	NULL
104	Sanket	12000	2
105	Ashwini	23000	1
106	Seema	30000	NULL

Dept		
Dept_No	D_Name	Location
1	IT	Pune
2	Admin	Mumbai
3	HR	Pune
4	SCM	Mumbai

FULL OUTER JOIN: Matching records from both tables, non-matching records from left table with null values for right table, non-matching records from right table with null values for left table

After full outer join:

```
SELECT * FROM employee LEFT JOIN dept
ON employee.Dept_No = dept.Dept_No;
UNION
SELECT * FROM employee RIGHT JOIN dept
ON employee.Dept_No = dept.Dept_No;
```

Eid	E_Name	Salary	Dept_No	Dept_No	Name	Location
101	Sanika	23000	1	1	IT	Pune
102	Pranoti	34000	2	2	Admin	Mumbai
104	Sanket	12000	2	2	Admin	Mumbai
105	Ashwini	23000	1	1	IT	Pune
103	Lokesh	45000	NULL	NULL	NULL	NULL
106	Seema	30000	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	3	HR	Pune
NULL	NULL	NULL	NULL	4	SCM	Mumbai

CROSS JOIN

Employee			
Eid	Name	Salary	**Dept_No**
101	Sanika	23000	1
102	Pranoti	34000	2
103	Lokesh	45000	NULL
104	Sanket	12000	2
105	Ashwini	23000	1
106	Seema	30000	NULL

Dept		
Dept_No	D_Name	Location
1	IT	Pune
2	Admin	Mumbai
3	HR	Pune
4	SCM	Mumbai

CROSS JOIN: Each record from one table will be mapped with each record from another table.

After cross join:

```
SELECT * FROM employee,dept:
```

Eid	E_Name	Salary	Dept_No	Dept_No	Name	Location
101	Sanika	23000	1	1	IT	Pune
101	Sanika	23000	1	2	Admin	Mumbai
101	Sanika	23000	1	3	HR	Pune
101	Sanika	23000	1	4	SCM	Mumbai
102	Pranoti	34000	2	1	IT	Pune
102	Pranoti	34000	2	2	Admin	Mumbai
102	Pranoti	34000	2	3	HR	Pune
102	Pranoti	34000	2	4	SCM	Mumbai
103	Lokesh	45000	NULL	1	IT	Pune
103	Lokesh	45000	NULL	2	Admin	Mumbai
103	Lokesh	45000	NULL	3	HR	Pune
103	Lokesh	45000	NULL	4	SCM	Mumbai
104	Sanket	12000	2	1	IT	Pune
104	Sanket	12000	2	2	Admin	Mumbai
104	Sanket	12000	2	3	HR	Pune
104	Sanket	12000	2	4	SCM	Mumbai
105	Ashwini	23000	1	1	IT	Pune
105	Ashwini	23000	1	2	Admin	Mumbai
105	Ashwini	23000	1	3	HR	Pune
105	Ashwini	23000	1	4	SCM	Mumbai
106	Seema	30000	NULL	1	IT	Pune
106	Seema	30000	NULL	2	Admin	Mumbai
106	Seema	30000	NULL	3	HR	Pune
106	Seema	30000	NULL	4	SCM	Mumbai

SELF JOIN

Eid	Name	Salary	Dept_no	M_id
101	Lilesh	45000	1	Null
102	Swapnil	34000	2	104
103	Vaishnavi	42000	1	101
104	Tejas	23000	2	Null
105	Arati	20000	1	104
106	Dipak	25000	1	101

SELF JOIN: A SELF JOIN is when a table is joined with itself. It's useful when the rows in the same table are related to each other — for example, an employee and their manager stored in the same table.

After self join:

```
SELECT  e.eid,e.name "Emp_name",m.name "Manager_name"
FROM manager m INNER JOIN manager p
ON m.mid = p.eid;
```

Eid	Emp_name	Manager_name
101	Swapnil	Tejas
103	Vaishanavi	Lilesh
105	Arati	Tejas
106	Dipak	Lilesh